# Sicong (Flora) Gu

sg4757@columbia.edu | flor-g.github.io | github.com/flor-g

*Columbia computer science student and linguist with experience in prompt design, multi-agent LLM workflows, and transformer models.*

## EDUCATION

**Columbia University**, Fu Foundation School of Engineering and Applied Science — **New York City, NY, USA**
*MS in Computer Science, expected completion June 2028* — *May 2026 - Present*

**University College London**, Faculty of Brain Sciences, Dept. of Linguistics — **London, UK**
*BA Linguistics (Hons)* — *September 2023 - 2026*

- Thesis: "Scalar Resolution under Local Computation: A Free-Energy Predictive-Coding Model of Implicature and Gradable Adjectives." Supervisor: Prof. Yasutada Sudo. Submitted September 2026; awaiting grade.
- Relevant coursework: Computational and Neural Linguistics; Formal Semantics; Generative Syntax; Discrete Mathematics.

## RESEARCH AND TECHNICAL EXPERIENCE

**Modelling Scalar Inference through Predictive Coding** — Dept. of Linguistics, UCL, **London, UK**
*Undergraduate Thesis Researcher | Supervisor: Prof. Yasutada Sudo* — *2025-2026*

- Built a custom PyTorch simulation with numerical quadrature, gradient-based inference, and closed-form stationary solutions.
- Established convergence and stability properties, and validated the implementation with 15 specification checks, including fixed-point identities, finite-difference gradients, Hessian analysis, grid refinement, and independent numerical verification.
- Evaluated the model against published gradable-adjective judgment data from Xiang et al. (2022), obtaining $R^2 = 0.993$ for the maximum-adjective class while identifying and analyzing systematic misfit in the minimum-adjective class.
- Demonstrated that the implemented architecture can produce scalar-implicature and endpoint-oriented interpretation patterns without explicit within-trial competition among alternative utterances, subject to stated model assumptions and boundary conditions.

**Transformer Representations and Psycholinguistic Prediction** — Dept. of Linguistics, UCL, **London, UK**
*Computational linguistics research project | Coauthor: Natasha Luo | Supervisor: Prof. Mario Giulianelli* — *2026*

- Tested five distance metrics across 13 GPT-2 Small layers to predict reading times and N400/P600 via incremental alternative sampling.
- Wrote zsh/Bash scripts to automate local GPT-2 inference and representation extraction with PyTorch's Apple Metal (MPS) backend; compared models through regression analyses in R, using change in adjusted $R^2$ beyond control variables.
- Found that Spearman rank distance most consistently improved reading-time prediction; output-level mixing improved mean model fit over embedding-level mixing, although the strongest single-layer model remained best overall.
- Presented at ULAB XVI, York, April 2026; forthcoming in the ULAB 2026 proceedings. Code: github.com/flor-g/PLIN0072_project_IAS

**Guangdong Laboratory of Artificial Intelligence and Digital Economy** — **Shenzhen, China**
*SWE Intern* — *July 2025 - September 2025*

- Redesigned a real-time clinical documentation assistant from a single-agent system into a multi-agent LLM workflow, with agents for transcript disambiguation, case drafting, and clinical review, and designed and optimized each agent's prompts and responsibilities.
- Refined prompts and edit-alignment constraints to reduce unsupported or reintroduced content across repeated updates.
- Implemented regex-based tracking of clinician edits across turns and a conditional final-step agent that preserves detected edits.
- Built a FastAPI microservice with Pydantic schemas, JSON and server-sent event streaming, and a Qwen-14B inference endpoint.
- Contributed the workflow to a clinical documentation system deployed at HKU-Shenzhen Hospital, with deployment led by others.

**Department of Linguistics, University College London** — **London, UK**
*Research Apprentice | PI: Prof. Klaus Abels* — *September 2024 - January 2025*

- Built an object-oriented Python stimulus compiler automating syntactic variants and wh-phrases for a 4 x 2 x 2 sluicing design.
- Curated stimuli, ran British National Corpus controls, and specified a mixed-effects analysis plan; ended before participant recruitment.

## SELECTED CONFERENCE PRESENTATIONS

- Gu, S. "Another Way to Wonky Worlds: Prior Precision Learning in Scalar Inference via the Free-Energy Principle." SLUgS, University of California, Berkeley, April 2026.
- Gu, S. "High-Absolutive Syntax in Mandarin Existential Sentences." CULC20, Cornell University, April 2026.

## TEACHING AND MENTORING

**Circustreams Studio Ltd.** — **London, UK**
*Cofounder and Course Instructor* — *September 2024 - Present*

- Designed, marketed, and taught two five-session introductory kintsugi courses that generated GBP 2,500 in pre-tax revenue.

**Resources for Autism** — **London, UK**
*Volunteer Mentor* — *September 2024 - March 2025*

- Provided weekly one-on-one mentorship to a neurodivergent adult facing long-term unemployment.

## TECHNICAL SKILLS

**Prompting and LLM tools:** Prompt engineering, multi-agent orchestration, JSON prompt templates, Hugging Face Transformers
**Programming:** Python, R, Java, JavaScript, Bash/zsh, HTML/CSS; C and Unix systems programming (in progress)
**ML and evaluation:** PyTorch, NumPy, Pandas, SciPy, Matplotlib, NLTK; mixed-effects regression, sensitivity analysis, experimental design
**Software and APIs:** FastAPI, Pydantic, OpenAI-compatible APIs, server-sent event streaming, requests/httpx, Git, LaTeX, Gorilla
**Languages:** English, Mandarin, Latin, Classical Greek, Classical Chinese